Week 8 · Class 1: Practice Exercises — ANSWER KEY

Sampling and Uncertainty & Statistical Significance

2025-12-31

1 Segment A — Sampling and Uncertainty

2 Non-AI Exercises

2.1 1. Populations vs. Samples

2.1.1 1.1 Multiple Choice: Basic Concepts

What is a population parameter?

a) An estimate from a sample
b) The true, fixed value in the complete population
c) A variable in a regression model
d) A hypothesis we want to test

Answer: b

2.1.2 1.2 Fill in the Blanks: Notation

Complete these definitions using proper notation:

1. μ represents the population mean
2. $\bar{x}$ represents the sample mean
3. σ represents the population standard deviation
4. s represents the sample standard deviation
5. n represents the sample size

2.1.3 1.3 True or False: Sampling Principles

Mark each statement as True (T) or False (F):

F All samples from the same population give identical results
T Larger samples generally give more precise estimates
T The sample mean is an unbiased estimator of the population mean
F We can eliminate sampling variation completely
T Different samples vary randomly but predictably **T** Random assignment (experiments) and random sampling (polls) both rely on chance producing predictable patterns

2.2 2. Sources of Sampling Variation

2.2.1 2.1 Match: Sources with Explanations

Match each source of variation with its explanation:

Sources:

a) Random selection
b) Population heterogeneity
c) Sample size limitations

Explanations:

1. Smaller samples are more susceptible to variation
2. Each sample includes different people by chance
3. The population itself is diverse

Matches: a = 2, b = 3, c = 1

2.2.2 2.2 Conceptual Question: Central Limit Theorem

Explain in your own words what the Central Limit Theorem tells us about sampling distributions:

Sample Answer: The Central Limit Theorem tells us that when we take many samples from a population and calculate the mean of each sample, those sample means will follow a normal (bell-shaped) distribution centered on the true population mean. This happens regardless of the shape of the original population distribution. This is important because it allows us to make predictions about how far our sample mean might be from the true population mean.

2.2.3 2.3 Multiple Choice: Law of Large Numbers

The Law of Large Numbers states that:

a) Large populations always have large means
b) With more observations, sample means get closer to the population mean
c) Larger samples always eliminate all error
d) Sample size doesn't affect precision

Answer: b

2.3 3. Standard Error and Margin of Error

2.3.1 3.1 Fill in the Blanks: Standard Error Formula

The standard error formula is:

$$SE = \frac{\sigma}{\sqrt{n}}$$

Complete the formula using: (population standard deviation), n (sample size)

2.3.2 3.2 True or False: Standard Error Concepts

Mark each statement as True (T) or False (F):

T Standard error decreases as sample size increases
T Standard error measures typical variation in sample estimates
F Margin of error is the same thing as standard error
F A standard error of 5% means we're 5% uncertain
T We can calculate standard error from a single sample

2.3.3 3.3 Calculation: Standard Error

A poll of n=400 voters finds support for a candidate. The population standard deviation is approximately 50 (for proportions measured as percentages). Calculate the standard error:

$$SE = \frac{50}{\sqrt{400}} = \frac{50}{20} = 2.5$$

2.3.4 3.4 Calculation: Margin of Error

Using the standard error from 3.3, calculate the margin of error (use the multiplier 2 for approximately 95% confidence):

$$MOE = 2 \times SE = 2 \times 2.5 = 5$$

2.4 4. Interpreting Uncertainty

2.4.1 4.1 Multiple Choice: Comparing Polls

Poll A (n=100) has SE = 5%. Poll B (n=900) has SE = 1.67%. Both estimate 52% support. Which is more reliable?

a) Poll A, because it's easier to conduct
b) Poll B, because it has a smaller standard error
c) They're equally reliable
d) Cannot determine without more information

Answer: b

2.4.2 4.2 Scenario: Close Election

A poll finds Candidate A at 48% and Candidate B at 52%, with a margin of error of ±4%. What can we conclude?

a) Candidate B will definitely win
b) The race is too close to call
c) Candidate A will definitely win
d) The poll is invalid

Answer: b

Explanation: With a margin of error of ±4%, Candidate A's true support could be anywhere from 44% to 52%, and Candidate B's true support could be anywhere from 48% to 56%. These ranges overlap significantly, meaning we cannot determine with confidence who is actually ahead.

2.4.3 4.3 Short Answer: Explaining Uncertainty

A news report says "Presidential approval is at 45% ±3%". In 2-3 sentences, explain what this means to someone who doesn't know statistics:

Sample Answer: This means that based on a poll, we estimate the president's approval rating is around 45%, but it could realistically be as low as 42% or as high as 48%. The ±3% represents the margin of error, which accounts for the fact that we only surveyed a sample of people rather than everyone in the country. We're fairly confident (about 95% sure) that the true approval rating falls somewhere within that range.

2.5 5. Sample Size, Precision, and the t-Distribution

2.5.1 5.1 Multiple Choice: Doubling Sample Size

To cut the standard error in half, you need to:

a) Double the sample size
b) Triple the sample size
c) Quadruple the sample size
d) Increase sample size by 50%

Answer: c

Explanation: Since SE = /√n, to cut SE in half, you need √n to double. If √n doubles, then n must be multiplied by 4 (quadrupled). For example, if n=100, SE 1/10. To get SE 1/20, you need n=400.

2.5.2 5.2 True or False: The t-Distribution

T The t-distribution has heavier tails than the normal distribution **T** We use it because the standard deviation is *estimated* from the sample **T** A sample of 25 has 24 degrees of freedom ($df = n - 1$) **T** As samples grow, the t-distribution approaches the normal **F** Small samples need a *smaller* critical value to reach significance

(Small samples need a *larger* critical value — e.g., ±2.571 at $df = 5$ vs. ±1.96 in large samples — so the bar for significance is *higher*.)

2.5.3 5.3 Multiple Choice: Which Distribution?

Answer: **c) The t-distribution with df = 14**

With unknown and estimated by s, the t-distribution accounts for that extra uncertainty; df = n − 1 = 14.

3 AI Exercises

Tips for Working with Claude:

- Ask for **R code using only tidyverse** (no other packages)
- Request **simple, focused answers** to your specific question—not complex analyses
- Ask Claude to **explain what the code is doing** since you're learning
- Avoid asking for visualizations or plots unless an exercise asks for one
- Include the output of `glimpse()` in your prompt so Claude knows your variable names

Example prompt: "Using tidyverse in R, calculate the mean and standard error for a variable. Keep the code simple and explain what each line does. Here is what my data looks like: [paste glimpse output]"

3.1 6. Computing Sample Statistics

3.1.1 6.1 Basic Calculations

```r
# Load libraries

# Load political social media data
social_media <- read_csv("data/political_text_ml.csv")
```

```
Rows: 500 Columns: 9
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (1): topic
dbl  (7): post_id, user_id, follower_count, contains_misinfo, engagements, s...
date (1): date

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# Calculate sample mean for engagements
mean_engagements <- mean(social_media$engagements, na.rm = TRUE)
cat("Mean engagements:", round(mean_engagements, 2), "\n")
```

```
Mean engagements: 182.57
```

```
# Calculate sample mean for sentiment
mean_sentiment <- mean(social_media$sentiment, na.rm = TRUE)
cat("Mean sentiment:", round(mean_sentiment, 3), "\n")
```

```
Mean sentiment: 0.022
```

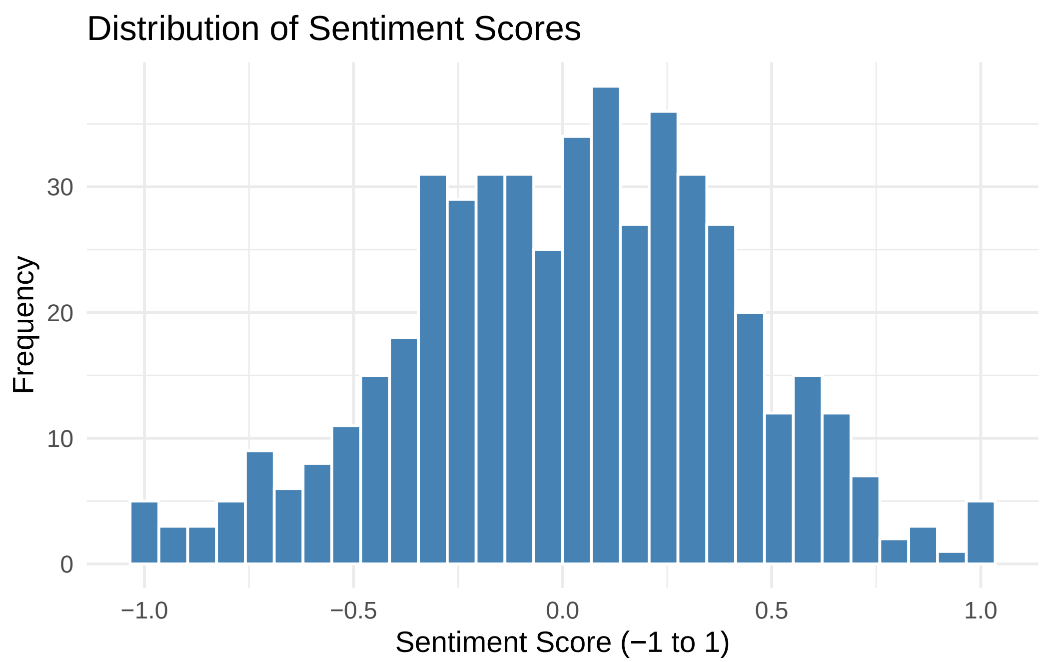
```
# Calculate sample standard deviation for both
sd_engagements <- sd(social_media$engagements, na.rm = TRUE)
cat("SD of engagements:", round(sd_engagements, 2), "\n")
```

```
SD of engagements: 736.85
```

```
sd_sentiment <- sd(social_media$sentiment, na.rm = TRUE)
cat("SD of sentiment:", round(sd_sentiment, 3), "\n")
```

```
SD of sentiment: 0.395
```

```
# Visualize sentiment distribution
ggplot(social_media, aes(x = sentiment)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  labs(
    title = "Distribution of Sentiment Scores",
    x = "Sentiment Score (-1 to 1)",
    y = "Frequency"
  ) +
  theme_minimal()
```



Interpretation: The sentiment distribution shows the range and frequency of sentiment scores across political social media posts. The mean sentiment indicates whether posts are generally positive or negative, while the standard deviation shows how much variation exists in sentiment across posts.

3.1.2 6.2 Comparing Two Groups

```
# Compare sentiment by verified status
verified_sentiment <- social_media %>%
  group_by(verified) %>%
  summarise(
    mean_sentiment = mean(sentiment, na.rm = TRUE),
    sd_sentiment = sd(sentiment, na.rm = TRUE),
    n = n()
  )

print(verified_sentiment)
```

```
# A tibble: 2 x 4
  verified mean_sentiment sd_sentiment     n
     <dbl>          <dbl>        <dbl> <int>
```

```
1         0          0.0260         0.392   455
2         1         -0.0198         0.435    45
```

```r
# Compare engagements by misinformation status
misinfo_engagement <- social_media %>%
  group_by(contains_misinfo) %>%
  summarise(
    mean_engagement = mean(engagements, na.rm = TRUE),
    sd_engagement = sd(engagements, na.rm = TRUE),
    n = n()
  )

print(misinfo_engagement)
```

```
# A tibble: 2 x 4
  contains_misinfo mean_engagement sd_engagement     n
             <dbl>           <dbl>         <dbl> <int>
1                0            154.          337.   414
2                1            320.         1616.    86
```

Interpretation: These comparisons show whether verified users have different sentiment patterns than unverified users, and whether posts containing misinformation receive more or less engagement than accurate posts. The standard deviations indicate how much variation exists within each group.

3.2 7. Standard Error Calculations

3.2.1 7.1 Computing Standard Error

```r
# Load policy outcomes data
policy_data <- read_csv("data/policy_outcomes_ml.csv")
```

```
Rows: 400 Columns: 8
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (2): recipient_country, sector
dbl (6): proj_id, start_year, commitment_usd_m, implementation_years, comple...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Filter to completed projects
completed_projects <- policy_data %>%
  filter(completed == 1, !is.na(eval_score))

# Calculate sample statistics
mean_eval <- mean(completed_projects$eval_score)
sd_eval <- sd(completed_projects$eval_score)
n_projects <- nrow(completed_projects)

# Calculate standard error
se_eval <- sd_eval / sqrt(n_projects)

cat("Sample mean evaluation score:", round(mean_eval, 2), "\n")
```

```
Sample mean evaluation score: 2.97
```

```r
cat("Sample SD:", round(sd_eval, 2), "\n")
```

```
Sample SD: 1.4
```

```r
cat("Sample size:", n_projects, "\n")
```

```
Sample size: 281
```

```r
cat("Standard error:", round(se_eval, 3), "\n")
```

```
Standard error: 0.083
```

Interpretation: The standard error tells us the typical amount by which our sample mean (of evaluation scores) might differ from the true population mean of all development projects. A smaller SE indicates more precision in our estimate.

3.2.2 7.2 Standard Error by Group

```r
# Calculate SE by sector
sector_stats <- policy_data %>%
  group_by(sector) %>%
  summarise(
    mean_funding = mean(commitment_usd_m, na.rm = TRUE),
    sd_funding = sd(commitment_usd_m, na.rm = TRUE),
    n = n(),
    se_funding = sd_funding / sqrt(n)
  ) %>%
  arrange(desc(se_funding))

print(sector_stats)
```

```
# A tibble: 5 x 5
  sector         mean_funding sd_funding     n se_funding
  <chr>                 <dbl>      <dbl> <int>      <dbl>
1 Governance             13.9       19.1    81       2.13
2 Health                 13.6       16.2    80       1.82
3 Education              12.1       14.0    74       1.62
4 Infrastructure         12.5       13.2    75       1.53
5 Other                  11.5       11.8    90       1.25
```

Interpretation: Sectors with smaller sample sizes will have larger standard errors, meaning we have more uncertainty about the true average funding for those sectors. Sectors with larger sample sizes provide more precise estimates of typical funding levels.

3.3 8. Margin of Error in Practice

3.3.1 8.1 Vote Share Analysis

```r
# Load election data
election_data <- read_csv("data/election_prediction_ml.csv")
```

```
Rows: 600 Columns: 6
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (2): state, county
dbl (4): FIPS, votes_dem_16, votes_gop_16, votes_total_16
```

```
i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Calculate Democratic vote share for each county
election_data <- election_data %>%
  mutate(dem_share = votes_dem_16 / votes_total_16)

# Calculate overall statistics
mean_dem_share <- mean(election_data$dem_share, na.rm = TRUE)
sd_dem_share <- sd(election_data$dem_share, na.rm = TRUE)
n_counties <- sum(!is.na(election_data$dem_share))

# Calculate standard error and margin of error
se_share <- sd_dem_share / sqrt(n_counties)
moe_share <- 2 * se_share

cat("Mean Democratic vote share:", round(mean_dem_share * 100, 2), "%\n")
```

```
Mean Democratic vote share: 48.17 %
```

```r
cat("Standard error:", round(se_share * 100, 2), "%\n")
```

```
Standard error: 0.46 %
```

```r
cat("Margin of error:", round(moe_share * 100, 2), "%\n")
```

```
Margin of error: 0.92 %
```

```r
cat("\nResult: ", round(mean_dem_share * 100, 2), "% ±",
    round(moe_share * 100, 2), "%\n")
```

```

Result:  48.17 % ± 0.92 %
```

Interpretation: This tells us the average Democratic vote share across all counties, along with a margin of error that reflects the sampling uncertainty if we consider these counties as a sample representing all possible counties.

3.3.2 8.2 State-Level Comparison

```r
# Calculate by state
state_stats <- election_data %>%
  group_by(state) %>%
  summarise(
    mean_dem_share = mean(dem_share, na.rm = TRUE),
    sd_dem_share = sd(dem_share, na.rm = TRUE),
    n_counties = n(),
    se_share = sd_dem_share / sqrt(n_counties),
    moe_share = 2 * se_share
  ) %>%
  arrange(desc(se_share))

# Show top and bottom states by SE
print("States with highest uncertainty (largest SE):")
```

```
[1] "States with highest uncertainty (largest SE):"
```

```r
print(head(state_stats, 5))
```

```
# A tibble: 5 x 6
  state       mean_dem_share sd_dem_share n_counties se_share moe_share
  <chr>                <dbl>        <dbl>      <int>    <dbl>     <dbl>
1 Idaho                0.452        0.129         20   0.0289    0.0578
2 Alabama              0.340        0.118         20   0.0264    0.0528
3 Iowa                 0.506        0.117         20   0.0262    0.0524
4 Hawaii               0.603        0.111         20   0.0249    0.0497
5 Connecticut          0.470        0.111         20   0.0249    0.0497
```

```r
print("\nStates with lowest uncertainty (smallest SE):")
```

```
[1] "\nStates with lowest uncertainty (smallest SE):"
```

```r
print(tail(state_stats, 5))
```

```
# A tibble: 5 x 6
  state         mean_dem_share sd_dem_share n_counties se_share moe_share
```

```
  <chr>                    <dbl>        <dbl>      <int>    <dbl>     <dbl>
1 Kansas                   0.483       0.0816         20   0.0182    0.0365
2 Minnesota                0.510       0.0793         20   0.0177    0.0354
3 Missouri                 0.501       0.0766         20   0.0171    0.0343
4 Maine                    0.492       0.0722         20   0.0161    0.0323
5 Massachusetts            0.605       0.0559         20   0.0125    0.0250
```

Interpretation: States with fewer counties will have larger standard errors and margins of error, meaning we have more uncertainty about the true state-level Democratic support. States with many counties provide more precise estimates. Additionally, states where counties are more similar to each other (lower SD) will have smaller standard errors than states with highly variable counties.

3.3.3 8.3 Interpreting Uncertainty

Sample Interpretation:

When comparing states, we need to consider both the point estimate (mean vote share) and the margin of error.

For example, if State A has 45% ± 2% and State B has 47% ± 5%, we cannot conclude with confidence that State B has higher Democratic support. The ranges overlap (State A: 43-47%, State B: 42-52%), so the true values could be quite similar or even reversed.

States with many counties give us more precise estimates (smaller margins of error), allowing for more confident comparisons. States with few counties or highly variable counties have larger margins of error, making comparisons less certain.

This is why understanding margin of error is crucial in interpreting election results - a small difference in point estimates might not be meaningful if the margins of error are large.

4 Segment B — What is Statistical Significance?

5 Specifying Hypotheses

5.1 Scenario 1: Voter Turnout

A political consultant claims that sending personalized text messages to voters increases turnout. To test this, you randomly assign 500 voters to receive text messages and 500 voters to a control group with no messages. You will measure turnout rates for both groups.

Null Hypothesis (H_0):

Text messages have no effect on voter turnout. The turnout rate for the text message group equals the turnout rate for the control group.

Or: $H_0 : \text{Turnout}_{\text{text}} = \text{Turnout}_{\text{control}}$

Alternative Hypothesis (H_1):

Text messages increase voter turnout. The turnout rate for the text message group is higher than the turnout rate for the control group.

Or: $H_1 : \text{Turnout}_{\text{text}} > \text{Turnout}_{\text{control}}$

5.2 Scenario 2: Presidential Approval

A news outlet wants to know whether the president's approval rating has changed from last month's level of 48%. They conduct a new poll of 1,000 registered voters.

Null Hypothesis (H_0):

The president's approval rating has not changed. Current approval = 48%.

Or: $H_0 : \text{Approval} = 48\%$

Alternative Hypothesis (H_1):

The president's approval rating has changed. Current approval   48%.

Or: $H_1 : \text{Approval} \neq 48\%$

Note: This is a two-tailed test because we're testing for any change (increase or decrease).

5.3 Scenario 3: Education and Political Knowledge

A researcher hypothesizes that college graduates have higher levels of political knowledge than those without college degrees. They survey 400 people (200 college graduates and 200 non-graduates) and measure their political knowledge on a 0-100 scale.

Null Hypothesis (H_0):

There is no difference in political knowledge between college graduates and non-graduates. Both groups have equal average political knowledge scores.

Or: $H_0 : \text{Knowledge}_{\text{college}} = \text{Knowledge}_{\text{non-college}}$

Alternative Hypothesis (H_1):

College graduates have higher political knowledge scores than non-graduates.

Or: $H_1 : \text{Knowledge}_{\text{college}} > \text{Knowledge}_{\text{non-college}}$

5.4 Scenario 4: Campaign Spending Effect

An incumbent politician believes that increased campaign spending will improve their vote share in the upcoming election. They want to test whether counties where they spend more money show higher support compared to last election's results.

Null Hypothesis (H_0):

Campaign spending has no effect on vote share. Vote share in high-spending counties equals vote share in low-spending counties (or compared to previous election).

Or: H_0: The relationship between spending and vote share is zero.

Alternative Hypothesis (H_1):

Increased campaign spending improves vote share. Counties with higher spending show higher vote share.

Or: H_1: There is a positive relationship between spending and vote share.

5.5 Scenario 5: Partisan News Consumption

A media researcher wants to test whether Republicans and Democrats differ in their average daily consumption of political news (measured in minutes per day). They survey 300 Republicans and 300 Democrats.

Null Hypothesis (H_0):

There is no difference in political news consumption between Republicans and Democrats. Both groups consume equal amounts of political news per day.

Or: $H_0 : \text{News}_{\text{Republican}} = \text{News}_{\text{Democrat}}$

Alternative Hypothesis (H_1):

Republicans and Democrats differ in their political news consumption.

Or: $H_1 : \text{News}_{\text{Republican}} \neq \text{News}_{\text{Democrat}}$

Note: This is a two-tailed test because we're testing for any difference, not a specific direction.

5.6 Scenario 6: Policy Support After Information

A policy advocacy group believes that providing voters with factual information about climate change will increase support for environmental policies. They randomly assign 600 participants to either receive an informational packet or not, then measure their policy support on a 0-10 scale.

Null Hypothesis (H_0):

Receiving information has no effect on policy support. Support levels are equal between the information group and the control group.

Or: $H_0 : \text{Support}_{\text{information}} = \text{Support}_{\text{control}}$

Alternative Hypothesis (H_1):

Receiving information increases policy support. The information group shows higher support than the control group.

Or: $H_1 : \text{Support}_{\text{information}} > \text{Support}_{\text{control}}$

6 Conceptual Questions

6.1 Question 1: Understanding the Logic

Explain in your own words why we test the null hypothesis rather than directly testing the alternative hypothesis.

Sample Answer:

We test the null hypothesis because it follows the logic of proof by contradiction. We can never definitively "prove" our theory is true, but we can show that the assumption of "no effect" is very unlikely given our data. By assuming the null hypothesis is true and then showing our observed data would be extremely unlikely under that assumption, we have evidence to reject the null. This is more scientifically rigorous than trying to directly prove our alternative hypothesis, because we're using probability and evidence rather than assertion.

Key Points: - Uses proof by contradiction logic - Can't prove alternative, only reject null - More rigorous scientific approach - Based on probability, not certainty

6.2 Question 2: Rejecting vs. Failing to Reject

What is the difference between "rejecting the null hypothesis" and "accepting the alternative hypothesis"? Are these the same thing? (You will need to look this up)

Sample Answer:

"Rejecting the null hypothesis" and "accepting the alternative hypothesis" are NOT the same thing. When we reject the null, we're saying "the data are very unlikely under the assumption of no effect," which gives us evidence FOR the alternative. However, we never truly "accept" or "prove" the alternative hypothesis - we only accumulate evidence against the null. This is an important distinction because:

1. There could be other explanations for our results
2. We're working with probability, not certainty
3. Future studies might find different results
4. Absence of evidence for the null is not the same as proof of the alternative

The proper language is: "We reject the null hypothesis" or "We fail to reject the null hypothesis," not "We accept the alternative." When we reject the null, we can say we have "evidence in support of" the alternative hypothesis, but we don't "accept" it as proven truth.

6.3 Question 3: Directionality

Consider these two alternative hypotheses:

- **A:** "The candidate's approval is different from 50%"
- **B:** "The candidate's approval is greater than 50%"

What is the key difference between these hypotheses? When might you choose one over the other?

Sample Answer:

The key difference is that Hypothesis A is **two-tailed** (tests for any difference in either direction), while Hypothesis B is **one-tailed** (tests for a difference in a specific direction).

Hypothesis A (two-tailed): Tests whether approval is either greater than OR less than 50%. We reject if we find strong evidence of any difference from 50%.

Hypothesis B (one-tailed): Tests only whether approval is greater than 50%. We reject only if we find strong evidence that approval is higher, not lower.

When to choose each:

- Use **two-tailed** (A) when you want to detect any change or difference, regardless of direction. This is appropriate when you're genuinely interested in whether there's any difference at all.
- Use **one-tailed** (B) when you have a specific directional prediction and only care about that direction. This is appropriate when the research question is specifically about whether something is higher (or lower).

Examples: - Testing if approval has *changed*: Use two-tailed (A) - Testing if a candidate has *majority support*: Use one-tailed (B) - Testing if two groups *differ*: Use two-tailed (A) - Testing if treatment *improves* outcomes: Use one-tailed (B)

Important note: Two-tailed tests are generally more conservative and are often preferred in research because they don't assume a direction ahead of time. One-tailed tests should only be used when you have a strong theoretical reason to expect an effect in only one direction.

6.4 Question 4: Interpreting a p-value

Sample Answer:

"If the president's true approval were really 50%, we would see a sample result as far from 50% as ours (46%) only about 0.6% of the time." The misreading avoided: the p-value is **not** the probability that the null hypothesis is true (nor a 99.4% probability that approval has dropped) — it describes the data assuming the null, not the null given the data.

6.5 Question 5: Statistical vs. Practical Significance

Sample Answer:

The result is statistically significant but almost certainly not important. With n = 40,000 the standard error is tiny, so even a 0.4-point shift on a 100-point scale is reliably *detectable*. The p-value answers "could a difference this size arise by chance if the intervention did nothing?" A policymaker cares about a different question: "is 0.4 points big enough to justify the cost?" Statistical significance depends on effect size *and* sample size; practical significance depends on the effect size alone, judged against real-world stakes. Both questions must be asked, and here they give different answers.